

The Stark-Zeeman Line-Shape Code

User Manual and Theoretical Guide

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Open-Source Coupled Stark-Zeeman Plasma Line-Shape Project

June 2026

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1 Introduction

StarkZee computes hydrogenic plasma line shapes when electric microfields and magnetic fields must be treated together rather than as independent corrections. The target problem is common in fusion spectroscopy, dense laboratory plasmas, and compact-object atmospheres: ionic microfields broaden and mix the emitter states through the Stark effect, while an imposed magnetic field splits and further mixes the same states through the Zeeman effect. When the two energy scales are comparable, a post-processed Stark profile with a Zeeman triplet added afterward is no longer sufficient.

The code therefore diagonalizes the Stark-Zeeman Hamiltonian directly at each sampled ionic microfield. It then averages the resulting polarized profiles over the microfield distribution and adds electron-impact broadening, optional ion dynamics through the Frequency Fluctuation Model (FFM), and post-processing convolutions such as thermal Doppler or instrumental broadening.

This manual is organized from use to derivation. Section 2 gives the computational workflow and package map. Section 3 derives the line-shape equations and the atomic, plasma, and numerical ingredients. Later sections summarize validation features, built-in comparison models, and the physical limits of the approximations.

2 Calculation Workflow and Package Map

The public entry point is the `LineProfile` class. A typical calculation builds a wavelength, frequency, or energy grid, calls `LineProfile.compute_profile(grid, grid_type)` — an alias of `compute_static_profile` — and then reads the total and polarization-resolved profiles from the resulting object. When ion dynamics matter (high density, low B , or high- n transitions), call `LineProfile.compute_ffm_profile(grid, grid_type)` instead: it requires `Ti_ev` to have been set on the constructor and populates the same result attributes via the Frequency Fluctuation Model. Internally, both high-level calls delegate the work through a small number of focused modules:

1. `line_profile.py` manages spectral grids, unit conversions, profile normalization, optional Doppler/slit convolutions, and user-facing result attributes.
2. `static_profile.py` performs the quasi-static microfield average. It creates the field-strength grid, the Gauss-Legendre orientation grid, and accumulates P_π , $P_{\sigma+}$, and $P_{\sigma-}$.
3. `radiator.py` constructs the field-free plus magnetic radiator Hamiltonian H_A and the transition dipole matrices in the uncoupled $|n, l, m_l, s, m_s\rangle$ basis.
4. `solve_starkzee()` forms the complete Hamiltonian $H = H_A + V_E$ at each microfield point and diagonalizes it as one matrix. The Stark term is not added perturbatively.
5. `broadening.py` evaluates electron-impact widths, including the frequency-dependent Griem-Baranger-Kolb (GBK) model and the ZEST-inspired variant.
6. `ffm.py` optionally replaces the static profile by a dynamic-ion profile using the Frequency Fluctuation Model, reachable from `LineProfile.compute_ffm_profile` or directly via `calculate_ffm_profile`.

7. `microfield.py` supplies Hooper, Holtsmark, Potekhin, and custom tabulated microfield distributions.
8. `models/` contains comparison models: `voigt`, `lomanowski`, `stehle_param`, `stehle`, and `rosato`.

The numerical calculation has one central loop: for each sampled microfield magnitude F and orientation $\mu = \cos \theta$, StarkZee builds $V_E(F, \mu)$, diagonalizes upper and lower manifolds, rotates the transition dipoles into the dressed-state bases, evaluates electron-broadened component profiles, and adds the weighted contribution to the three polarization channels. This is the organizing idea behind the equations that follow.

3 Physics and Numerical Formulation

3.1 Liouville Space Representation

The spectral line-shape intensity $I(\omega)$ of emitted radiation is related to the Fourier transform of the dipole autocorrelation function $C(t)$ by [2]:

$$I(\omega) = \frac{1}{\pi} \operatorname{Re} \int_0^\infty C(t) e^{i\omega t} dt \quad (1)$$

In Liouville space notation, the autocorrelation function is a trace over the quantum emitter states:

$$C(t) = \langle \langle \vec{d}^* | U(t) | \vec{d} \rho_0 \rangle \rangle \quad (2)$$

where $\vec{d}$ is the electric transition dipole operator, ρ_0 is the equilibrium density matrix of the initial manifold, and $U(t)$ is the time-evolution propagator.

In the anti-symmetric subspace (ground-to-excited manifold transitions in the no-quenching approximation), the Liouvillian operator L is constructed from the excited-state Hamiltonian H_u and ground-state Hamiltonian H_l :

$$L = \frac{1}{\hbar} (H_u \otimes \hat{I}^d - \hat{I} \otimes H_l^d) \quad (3)$$

The diagonal elements of L give the field-free transition frequencies $\omega_{ij} = (E_i^u - E_j^l)/\hbar$; off-diagonal elements encode the Stark and Zeeman couplings between dressed states.

From propagator to resolvent. The time-evolution propagator is generated by L :

$$U(t) = e^{-iLt} \quad (4)$$

Substituting into Eq. (2) gives the explicit correlation function

$$C(t) = \langle \langle \vec{d}^* | e^{-iLt} | \vec{d} \rho_0 \rangle \rangle \quad (5)$$

Inserting this into Eq. (1) and evaluating the one-sided Fourier transform converts the time-domain exponential into a frequency-domain resolvent:

$$I(\omega) = \frac{1}{\pi} \operatorname{Re} \langle \langle \vec{d}^* | (\omega \hat{I} - L)^{-1} | \vec{d} \rho_0 \rangle \rangle \quad (6)$$

Fast electron collisions are encoded as an additional imaginary damping operator $\Phi(\omega)$ in the same Liouville space (Section 3.8), shifting $L \rightarrow L + i\Phi(\omega)$:

$$I(\omega) = \frac{1}{\pi} \operatorname{Re} \langle \langle \vec{d}^* | (\omega \hat{I} - L - i\Phi(\omega))^{-1} | \vec{d} \rho_0 \rangle \rangle \quad (7)$$

Field-dependent Liouvillian and static profile. Under the quasi-static ion approximation the radiator is exposed to a static ionic microfield $\vec{F}$ at angle θ to $\vec{B}$. Both manifold Hamiltonians H_u and H_l then depend on the field magnitude F and orientation $\mu = \cos \theta$ through the Stark perturbation V_E (Section 3.4), making the Liouvillian parametrically field-dependent:

$$L(F, \mu) = \frac{1}{\hbar} \left[H_u(F, \mu) \otimes \hat{I}^d - \hat{I} \otimes H_l^d(F, \mu) \right], \quad H(F, \mu) = H_A + V_E(F, \mu) \quad (8)$$

where H_A is the field-free magnetic Hamiltonian (Section 3.2) and $V_E(F, \mu)$ is the linear Stark interaction (Section 3.4). Equations (7) and (8) together give the field-dependent q -polarized profile at a given configuration (F, μ) :

$$I_q(\omega, F, \mu) = \frac{1}{\pi} \text{Re} \langle d_q^* | (\omega \hat{I} - L(F, \mu) - i \Phi(\omega))^{-1} | d_q \rho_0 \rangle \rangle \quad (9)$$

The total q -polarized profile is then the average over the ionic microfield distribution $W(F)$ and orientation:

$$I_q(\omega) = \frac{1}{2} \int_0^\infty \int_{-1}^1 W(F) I_q(\omega, F, \mu) d\mu dF \quad (10)$$

Because $I_q(\omega, F, \mu) = I_q(\omega, F, -\mu)$ (the Hamiltonian is even in μ via $F_z = F\mu$ and $F_x = F\sqrt{1-\mu^2}$), the integral over $[-1, 1]$ reduces to twice the $[0, 1]$ range, discretized by Gauss-Legendre quadrature. The observed intensity at angle α between the line of sight and $\vec{B}$ is:

$$I(\omega, \alpha) = I_\pi(\omega) \sin^2 \alpha + \frac{1}{2} (I_{\sigma^+}(\omega) + I_{\sigma^-}(\omega)) (1 + \cos^2 \alpha) \quad (11)$$

Sections 3.2–3.8 detail each ingredient of Eq. (9); Section 3.10 extends this framework to include ion dynamics via the FFM.

3.2 Radiator Hamiltonian in Magnetic Fields

For each principal quantum number shell n , the atomic Hamiltonian H_A is constructed in the canonical uncoupled hydrogenic basis $|n, l, m_l, s=\frac{1}{2}, m_s\rangle$ of dimension $2n^2$. Basis states are ordered by an outer loop over $l \in [0, n-1]$, an intermediate loop over $m_l \in [-l, l]$, and an inner loop over $m_s \in \{+\frac{1}{2}, -\frac{1}{2}\}$. Five physical contributions are assembled into a single matrix [3]:

$$H_A = H_0 + H_{\text{SO}} + H_{\text{FS}} + H_Z^{(1)} + H_Z^{(2)} \quad (12)$$

3.2.1 Unperturbed Hydrogenic Energy (H_0)

The diagonal unperturbed energy is corrected for the finite nuclear mass:

$$\langle n, l, m_l, m_s | H_0 | n, l, m_l, m_s \rangle = -\frac{Z^2 \text{Ry}_{\text{red}}}{n^2}, \quad \text{Ry}_{\text{red}} = \text{Ry}_\infty \frac{M_{\text{nuc}}}{m_e + M_{\text{nuc}}} \quad (13)$$

where M_{nuc} is the nuclear mass of the emitting isotope (fixed by its atomic mass number) and Ry_{red} the corresponding reduced-mass Rydberg constant. When `use_empirical_data=True` this analytic diagonal is replaced by NIST tabulated energies (Section 3.3).

3.2.2 Spin-Orbit Coupling (H_{SO})

The spin-orbit interaction $H_{\text{SO}} = \zeta_{nl} \vec{L} \cdot \vec{S}$ couples orbital and spin angular momenta. The coupling constant for $l > 0$ is:

$$\zeta_{nl} = \frac{Z^4 \alpha^2 \text{Ry}_\infty}{n^3 l (l + \frac{1}{2}) (l + 1)} \quad (14)$$

Using the ladder decomposition $\vec{L} \cdot \vec{S} = L_z S_z + \frac{1}{2}(L_+ S_- + L_- S_+)$, the matrix elements within a given l -subshell are:

- **Diagonal** ($\Delta m_l = 0, \Delta m_s = 0$):

$$\langle l, m_l, m_s | H_{SO} | l, m_l, m_s \rangle = \xi_{nl} m_l m_s \quad (15)$$

- **Off-diagonal** ($\Delta m_l = \pm 1, \Delta m_s = \mp 1$): for $s = \frac{1}{2}$ the spin ladder factor equals 1 for all allowed transitions:

$$\langle l, m_l \pm 1, m_s \mp 1 | H_{SO} | l, m_l, m_s \rangle = \frac{1}{2} \xi_{nl} \sqrt{l(l+1) - m_l(m_l \pm 1)} \quad (16)$$

3.2.3 Fine-Structure Relativistic Corrections (H_{FS})

To restore the Dirac degeneracy (e.g. $2s_{1/2}$ and $2p_{1/2}$), the mass-velocity and Darwin terms are added [3]. These are diagonal in $|n, l, m_l, m_s\rangle$ and independent of m_l and m_s :

$$\langle l | H_{FS} | l \rangle = \begin{cases} -A_{FS} \left(n - \frac{3}{4} \right) & l = 0 \\ -A_{FS} \left(\frac{n}{l + \frac{1}{2}} - \frac{3}{4} \right) & l > 0 \end{cases} \quad (17)$$

where $A_{FS} = Z^4 \alpha^2 \text{Ry}_\infty / n^4$. The algebraic identity $H_{SO} + H_{FS}$ reproduces the exact Dirac fine-structure eigenvalues for both $j = l \pm \frac{1}{2}$.

3.2.4 Linear Zeeman Effect ($H_Z^{(1)}$)

$$\langle m_l, m_s | H_Z^{(1)} | m_l, m_s \rangle = \mu_B B (m_l + g_s m_s) \quad (18)$$

where $\mu_B \approx 5.788 \times 10^{-5} \text{ eV/T}$ and $g_s \approx 2.0023192$.

3.2.5 Quadratic (Diamagnetic) Zeeman Effect ($H_Z^{(2)}$)

$$H_Z^{(2)} = \frac{e^2 B^2}{8m_e} r^2 \sin^2 \theta \quad (19)$$

This operator couples states with $\Delta l = 0$ and $\Delta l = \pm 2$. The matrix elements factor into radial and angular parts:

$$\langle l_1, m_l | H_Z^{(2)} | l_2, m_l \rangle = \left(\frac{e^2 B^2 a_0^2}{8m_e} \right) \langle n, l_1 | r^2 | n, l_2 \rangle \langle l_1, m_l | \sin^2 \theta | l_2, m_l \rangle \quad (20)$$

1. Radial integrals:

- Diagonal ($l_1 = l_2 = l$): $\langle n, l | r^2 | n, l \rangle = \frac{n^2}{2Z^2} [5n^2 + 1 - 3l(l+1)] [a_0^2]$
- Off-diagonal ($l_2 = l_1 \pm 2$): evaluated numerically via $\int_0^\infty R_{nl_1} r^4 R_{nl_2} dr$, cached with `lru_cache`.

2. Angular integrals:

- Diagonal: for $l = 0$, $\langle \sin^2 \theta \rangle = 2/3$; for $l > 0$,

$$\langle l, m_l | \sin^2 \theta | l, m_l \rangle = 1 - \frac{2l^2 + 2l - 1 - 2m_l^2}{(2l-1)(2l+3)} \quad (21)$$

- Off-diagonal ($l_2 = l_1 + 2, l_{\text{low}} = \min(l_1, l_2)$): the constant term in $1 - \cos^2 \theta$ vanishes for orthogonal harmonics, leaving:

$$\langle l_1, m_l | \sin^2 \theta | l_2, m_l \rangle = -\sqrt{\frac{[(l_{\text{low}} + 1)^2 - m_l^2] [(l_{\text{low}} + 2)^2 - m_l^2]}{(2l_{\text{low}} + 1)(2l_{\text{low}} + 3)^2(2l_{\text{low}} + 5)}} \quad (22)$$

3.3 Empirical Field-Free Energies

The analytic diagonal $H_0 = -Z^2 \text{Ry}_{\text{red}}/n^2$, together with the spin-orbit and fine-structure terms, reproduces the field-free level positions only to the accuracy of the hydrogenic Dirac formula. For quantitative line centers, StarkZee can instead inject *measured* field-free energies through the `use_empirical_data=True` flag (with the element symbol `atom`). The values are read from a tabulated database (`atomic_data.load_levels`) holding NIST level energies [13] in wavenumbers [cm^{-1}], resolved by (n, l, j) . The database is stored in `starkzee/data/atomic_levels.json`; it currently holds hydrogen ("H", $n \leq 8$), deuterium ("D", $n \leq 6$), and tritium ("T", $n \leq 3$). Each top-level key is an atom symbol; the value is an object with two sections: "fine_structure_true" (entries with fields `n`, `l`, `j`, `energy`) and "fine_structure_false" (entries with fields `n`, `energy`). All energies are in cm^{-1} above the ground state (NIST vacuum values). Additional atoms can be supported by adding the corresponding entry to this file. To refresh the bundled data from the NIST ASD levels query, run `python scripts/update_nist_levels.py H -spectrum "H I"`. The updater is a development tool: it writes the local JSON database and checks that each requested fine-structure shell contains all hydrogenic (l, j) states before replacing the data; it also keeps NIST rows marked uncertain (a trailing ? on the energy value) rather than silently dropping them. The empirical Hamiltonian is kept in cm^{-1} throughout this code path so that callers can work entirely in wavenumber units; all Zeeman contributions are also converted from eV to cm^{-1} before being added, keeping every term on the same scale.

Beyond the bare Hamiltonian. `use_empirical_data/atom` are forwarded end-to-end by `calculate_static_profile`, `calculate_ffm_profile`, and the corresponding `LineProfile.compute_static_profile/compute_ffm_profile` methods. When active, the Stark matrices M_z, M_x (built once in eV per (n, Z) , Section 3.4) are rescaled to $\text{cm}^{-1}/(\text{V/m})$ so that $V_E = F_z M_z + F_x M_x$ stays unit-consistent with the cm^{-1} empirical Hamiltonian, and the diagonal-centering reference used to condition the eigensolve becomes the mean of the empirical levels rather than the analytic E_n ; the gross-structure line center E_0 used for GBK detunings and for the returned transition energies is derived from that same empirical reference ($E_0 = E_{n_u}^{\text{emp}} - E_{n_l}^{\text{emp}}$ converted back to eV), so absolute energies stay correct throughout. **Isotope caveat:** `atomic_levels.json` tabulates each isotope's own levels (H, D, T) but applies no further per-isotope correction — pass `atom` matching the emitting species (e.g. `atom='D'` with `species='D'`) to get the true isotope-shifted line center; combining empirical data for one isotope with a different emitting species reproduces that *other* isotope's line center instead.

Because the field-free Hamiltonian is degenerate (the Dirac formula makes $2s_{1/2}$ and $2p_{1/2}$ coincide), the empirical energies cannot simply be written on the diagonal of the uncoupled $|n, l, m_l, m_s\rangle$ basis: `numpy.linalg.eigh` is free to mix the degenerate l states arbitrarily. StarkZee therefore:

1. diagonalizes a degeneracy-broken field-free Hamiltonian (unperturbed energy + spin-orbit only, *omitting* the mass-velocity/Darwin term so that l remains a good label), yielding eigenvectors V ;
2. labels each coupled eigenstate k by its dominant orbital component l and its total angular momentum $j = l \pm \frac{1}{2}$, the branch being set by the sign of $\langle \vec{L} \cdot \vec{S} \rangle = (E_k -$

$E_n)/\xi_{nl}$;

3. assigns the tabulated energy $D_k = E^{\text{emp}}(l, j)$ [cm^{-1}] to each eigenstate and reconstructs the field-free Hamiltonian as

$$H_0^{\text{emp}} = V \text{diag}(D) V^\dagger \quad [\text{cm}^{-1}]. \quad (23)$$

The Zeeman terms ($\mu_B B(m_l + g_s m_s)$ and the diamagnetic correction) are converted from eV to cm^{-1} and added on top of H_0^{emp} . Since the tabulated values are absolute level energies (ground state at 0), transition wavenumbers follow directly as differences $E_u - E_l$ and reproduce the observed NIST Lyman/Balmer line centers.

3.4 Full Electron-Radiator Hamiltonian

Under the quasi-static ion approximation, the radiator is subjected to a constant electric microfield $\vec{F}$ at an angle θ relative to the magnetic field $\vec{B}$. The microfield is decomposed into longitudinal ($F_z = F \cos \theta$) and transverse ($F_x = F \sin \theta$) components. The Stark interaction is

$$V_E = -e \vec{r} \cdot \vec{F} = -e(zF_z + xF_x) = -\hat{d} \cdot \vec{F} \quad (24)$$

where $\hat{d} = e\vec{r}$ is the electric dipole operator. The full electron-radiator Hamiltonian is

$$H = H_A + V_E = H_0 + V_{\text{SO}} + H_Z^{(1)} + H_Z^{(2)} + V_E \quad (25)$$

H_A and V_E are assembled into a single matrix and diagonalized *as a whole* by `numpy.linalg.eigh` at every microfield quadrature point (`solve_starkzee`). The Stark interaction is **not** treated perturbatively: there is no expansion in powers of V_E . This exact treatment is essential when the Stark energy $eF\langle r \rangle_n$ is comparable to the fine-structure or Zeeman splittings, yielding the Stark-dressed eigenstates $|k\rangle$ and transition frequencies ω_k .

3.4.1 Stark Perturbation Matrix Elements

The selection rule for the within-shell Stark coupling is $\Delta n = 0$, $|l_i - l_j| = 1$, $\Delta m_s = 0$. In energy units [eV]:

$$\langle n, l_i, m_{l,i} | V_E | n, l_j, m_{l,j} \rangle = -\delta_{m_{s,i}, m_{s,j}} \langle n, l_i | r | n, l_j \rangle \langle l_i, m_{l,i} | (zF_z + xF_x) / r | l_j, m_{l,j} \rangle \quad (26)$$

Radial matrix element. For $l = \max(l_i, l_j)$, Gordon's formula gives the within-shell radial dipole element exactly:

$$\langle n, l | r | n, l-1 \rangle = \frac{3n}{2Z} \sqrt{n^2 - l^2} \quad [a_0] \quad (27)$$

Angular matrix elements. The angular factor is the spherical dipole tensor, written $z/r = T_0^{(1)}$ and $x/r = (T_{-1}^{(1)} + T_{+1}^{(1)})/\sqrt{2}$. Its non-zero matrix elements (with $l = \max(l_1, l_2)$) are:

$$\langle l, m_l | T_0^{(1)} | l+1, m_l \rangle = \sqrt{\frac{(l+1)^2 - m_l^2}{(2l+1)(2l+3)}} \quad (q=0, \Delta m_l=0) \quad (28)$$

$$\langle l_1, m_{l,1} | T_{+1}^{(1)} | l_2, m_{l,2} \rangle = \begin{cases} +\sqrt{\frac{(l-m_{l,2})(l-m_{l,2}-1)}{2(2l-1)(2l+1)}} & l_1 = l_2 - 1 \\ -\sqrt{\frac{(l+m_{l,2})(l+m_{l,2}+1)}{2(2l-1)(2l+1)}} & l_1 = l_2 + 1 \end{cases} \quad (q=+1, \Delta m_l=-1) \quad (29)$$

$$\langle l_1, m_{l,1} | T_{-1}^{(1)} | l_2, m_{l,2} \rangle = \begin{cases} -\sqrt{\frac{(l+m_{l,2})(l+m_{l,2}-1)}{2(2l-1)(2l+1)}} & l_1 = l_2 - 1 \\ +\sqrt{\frac{(l-m_{l,2})(l-m_{l,2}+1)}{2(2l-1)(2l+1)}} & l_1 = l_2 + 1 \end{cases} \quad (q = -1, \Delta m_l = +1) \quad (30)$$

These same three matrix elements reappear unchanged in the transition dipole operator (Section 3.5), since both the Stark coupling and the radiative dipole are built from the rank-1 tensor $T_q^{(1)}$. The Stark matrices M_z and M_x [eV/(V/m)] are precomputed once per (n, Z) and cached (`_stark_templates`), so that $V_E = F_z M_z + F_x M_x$ is formed at each quadrature point by a simple scalar multiplication.

3.5 Transition Dipole Operator

The transition dipole matrices D_q ($q \in \{0, \pm 1\}$) are built in the uncoupled basis $|n, l, m_l, s, m_s\rangle$. Selection rules require $\Delta l = \pm 1$, $\Delta m_l = q$, and $\Delta m_s = 0$:

$$\langle n_l, l_l, m_{l,l}, m_{s,l} | D_q | n_u, l_u, m_{l,u}, m_{s,u} \rangle = \delta_{m_{s,l}, m_{s,u}} \langle n_l, l_l | r | n_u, l_u \rangle \langle l_l, m_{l,l} | T_q^{(1)} | l_u, m_{l,u} \rangle \quad (31)$$

3.5.1 Angular Dipole Matrix Elements

The angular factors $\langle l_l, m_{l,l} | T_q^{(1)} | l_u, m_{l,u} \rangle$ follow from the Wigner-Eckart theorem [5]. They are exactly the rank-1 spherical-tensor matrix elements already written for the Stark coupling in Eqs. (28)–(30), now evaluated between the lower and upper manifolds: $q = 0$ gives the π component ($\Delta m_l = 0$), $q = +1$ the σ^- component ($\Delta m_l = -1$), and $q = -1$ the σ^+ component ($\Delta m_l = +1$).

3.5.2 Hydrogenic Radial Dipole Integrals (Gordon's Formula)

For $n_u \neq n_l$, the inter-shell radial elements $\langle n_l, l_l | r | n_u, l_u \rangle$ are evaluated exactly using Gordon's (1929) analytical formula [4]:

$$\begin{aligned} \langle n_l, l_l | r | n_u, l_u \rangle &= \frac{1}{Z} \frac{(-1)^{n_l-L}}{4(2L-1)!} \sqrt{\frac{(n_u+L)!(n_l+L-1)!}{(n_u-L-1)!(n_l-L)!}} \\ &\quad \times \frac{(4n_u n_l)^{L+1} (n_u - n_l)^{n_u+n_l-2L-2}}{(n_u + n_l)^{n_u+n_l}} \left[F_1 - \left(\frac{n_u - n_l}{n_u + n_l} \right)^2 F_2 \right] \end{aligned} \quad (32)$$

where $L = \max(l_u, l_l)$, and F_1, F_2 are terminating Gauss hypergeometric functions:

$$F_1 = {}_2F_1 \left(-n_u + L + 1, -n_l + L, 2L, -\frac{4n_u n_l}{(n_u - n_l)^2} \right) \quad (33)$$

$$F_2 = {}_2F_1 \left(-n_u + L - 1, -n_l + L, 2L, -\frac{4n_u n_l}{(n_u - n_l)^2} \right) \quad (34)$$

Because the first argument in both hypergeometric functions is a non-positive integer, each series terminates as a finite sum:

$${}_2F_1(-m, b, c, x) = \sum_{k=0}^m \frac{(-m)_k (b)_k}{(c)_k k!} x^k \quad (35)$$

where $(-m)_k$ is the Pochhammer symbol. This exact evaluation carries no integration cutoff error and is cached for reuse.

3.6 Diagonalization and Transition Dipole Basis Rotation

At each quadrature point (F, μ) the total Hamiltonians for both manifolds are:

$$H_u(F, \mu) = H_{A,u} + V_{E,u}(F_z, F_x), \quad H_l(F, \mu) = H_{A,l} + V_{E,l}(F_z, F_x) \quad (36)$$

Diagonalization via `numpy.linalg.eigh` gives eigenvectors V_u, V_l and eigenvalues E_u^p, E_l^k :

$$V_u^\dagger H_u V_u = \text{diag}(E_u^p), \quad V_l^\dagger H_l V_l = \text{diag}(E_l^k) \quad (37)$$

The transition energy for each dressed-state pair ($p \rightarrow k$) is $E_{pk} = E_u^p - E_l^k$. The uncoupled dipole matrices D_q are rotated into the mixed eigenstate bases:

$$D'_q = V_l^\dagger D_q V_u, \quad S_q(p, k) = |D'_q(k, p)|^2 \quad (38)$$

The weight $S_q(p, k)$ and energy E_{pk} together define one Stark-Dressed Transition (SDT). The static profile at field configuration (F, μ) — Eq. (9) — is then realized in practice as a sum over all SDTs weighted by the quadrature weight W_{ij} :

$$I_q(\omega, F, \mu) = \sum_{p,k} W_{ij} S_q(p, k) \frac{\gamma_e / \pi}{(\omega - E_{pk})^2 + \gamma_e^2} \quad (39)$$

where γ_e is the electron-impact half-width of Section 3.8.

3.7 Plasma Microfield Distributions

The electric microfield is averaged over a probability distribution $W(F)$. The presence of $\vec{B}$ breaks spherical symmetry, so the integration must be performed over both the field magnitude $F = |\vec{F}|$ and the orientation $\mu = \cos \theta$ between $\vec{F}$ and $\vec{B}$.

3.7.1 The Hooper Screened Microfield Distribution

To account for Debye screening of the ion Coulomb fields by surrounding plasma electrons, StarkZee implements a Hooper-like screened microfield distribution [11]:

$$W(\beta, a) = \frac{2\beta}{\pi} \int_0^\infty y \sin(\beta y) T(y, a) dy \quad (40)$$

Provenance note. This is the analytic screened characteristic-function ansatz below — it reduces to Holtsmark at $a = 0$ and approximates the Debye suppression Hooper's low-frequency-component theory predicts — not an interpolation of Hooper's own tabulated numerical results. For quantitative work at strong screening ($a \gtrsim 1$) prefer a tabulated distribution via `custom_table_path`, or the Potekhin distribution below. where:

- $\beta = F/F_0$ is the reduced electric field strength.
- F_0 is the **Holtsmark normal field strength**, the Coulomb field of a single perturber at the mean inter-particle distance r_e :

$$F_0 = \frac{e}{4\pi\epsilon_0 r_e^2}, \quad r_e = \left(\frac{3}{4\pi N_e} \right)^{1/3} \quad (41)$$

- $a = r_e/\lambda_D$ is the Debye screening parameter, where λ_D is the multi-species Debye length:

$$\lambda_D = \sqrt{\frac{\epsilon_0 k_B T_e}{N_e e^2 (1 + \sum_i X_i Z_i^2)}} \quad (42)$$

with $X_i = N_i/N_e$ the fractional ion concentration.

- $T(y, a) = \exp(-y^{3/2}S(y, a))$ is the screened characteristic function, with Debye screening factor:

$$S(y, a) = \left(1 + \frac{1.5 a^2}{y^2}\right)^{-3/4} \quad (43)$$

Setting $a = 0$ yields $S = 1$ and $T = \exp(-y^{3/2})$, recovering the unscreened **Holtmark distribution** $W_H(\beta)$ [10].

3.7.2 The Potekhin Microfield Distribution

The analytical fits of Potekhin et al. (2002) [12] describe the cumulative distribution $Q(\beta)$ and probability density $P(\beta)$ for both screened ($s > 0$) and unscreened ($s = 0$) potentials, for neutral and charged radiators, incorporating the ion-ion coupling parameter Γ :

$$Q(\beta) = \frac{q_0\beta^3 - 1.33\beta^{9/2} + \beta^6}{q_1 + q_2\beta^2 + q_3\beta^3 - \frac{1}{3}\beta^{9/2} + \beta^6} \quad (s = 0) \quad (44)$$

$$P(\beta) \approx \beta^2 S_N \left[A e^{-a\beta^\alpha} + B e^{-b\beta^\gamma} + \frac{e^{-\Gamma\beta^{1/2}}}{1 + c\beta^{9/2}} \right] \quad (s > 0) \quad (45)$$

where $q_n, A, a, \alpha, B, b, \gamma, c$ are tabulated functions of Γ and s , and S_N is the normalization constant.

3.7.3 Quadrature Discretization

The double integral $\int \int W(F, \mu) dF d\mu$ (Eq. 10) is discretized as follows:

1. **Field magnitude F :** A uniform grid of N_F points (`num_f`) in reduced field space $\beta_i \in [0, \beta_{\max}]$ (`max_beta`, default $\beta_{\max} = 10$):

$$F_i = \beta_i F_0, \quad W_{F,i} = W(\beta_i, a) \Delta\beta, \quad \sum_i W_{F,i} = 1 \quad (46)$$

The core of $W(\beta)$ sits below $\beta \approx 5-8$, but the Holtmark tail decays only as $\beta^{-5/2}$: about 3 % of the probability lies beyond $\beta = 10$, and truncating it (the weights above are renormalized to sum to 1) removes exactly the strong-field configurations that build the quasi-static far wings of the line. Increase `max_beta` (together with `num_f`, since the same point count must now cover a wider range) for far-wing studies; the default is adequate for line-core work.

2. **Field orientation μ :** N_μ -point **Gauss-Legendre quadrature** over $[0, 1]$. Legendre roots $x_j \in [-1, 1]$ and weights w_j are mapped as:

$$\mu_j = \frac{x_j + 1}{2}, \quad W_{\mu,j} = \frac{w_j}{2} \quad (47)$$

3. **Combined weight:** $W_{ij} = W_{F,i} W_{\mu,j}$. Points with $W_{ij} < 10^{-15}$ are skipped.

so that at each active point the field decomposes into the longitudinal and transverse components seen by the radiator,

$$F_{z,ij} = F_i \mu_j, \quad F_{x,ij} = F_i \sqrt{1 - \mu_j^2}, \quad (48)$$

which are passed directly to the Stark template (Section 3.4). The weight $W(\beta_i, a)$ is drawn from whichever distribution the user selects — Holtmark, Hooper, or Potekhin (above) — while the orientation quadrature is identical in every case.

3.8 Electron Impact Broadening

Fast-moving electrons are treated in the *impact* (completed-collision) approximation: each collision is completed in a time short compared to both the inverse linewidth and the mean inter-collision time. In this limit, successive collisions are statistically independent and their net effect on the line shape enters through a damping operator $\Phi(\omega)$ in the Liouville-space resolvent (Eq. 7).

3.8.1 Liouville-space structure of $\Phi(\omega)$ and connection to $\langle r^2 \rangle$

In the no-quenching approximation (electron collisions do not mix the upper and lower radiating manifolds), the operator Φ in Liouville space factorizes as a sum of independent contributions from each manifold:

$$\Phi(\omega) = \frac{1}{2} (\Phi_u(\omega) \otimes \hat{I}^d + \hat{I} \otimes \Phi_l^d(\omega)) \quad (49)$$

where Φ_u and Φ_l are self-energy operators acting within the upper (n_u) and lower (n_l) manifolds respectively.

The GBK semi-classical model [9, 8] evaluates Φ_n from the leading long-range term of the electron–radiator interaction, which is the dipole–dipole coupling $V_{ee} \propto r_{\text{atom}}^2 / r_e^3$. Averaging over the Maxwell–Boltzmann distribution of electron impact parameters and velocities, this yields a self-energy proportional to the operator r_n^2 within shell n :

$$\Phi_n(\omega) = W_0(\omega) r_n^2 [C_n + G(\omega)] \quad (50)$$

where r_n^2 is the mean-square-displacement operator within shell n , W_0 is the density-temperature prefactor (below), and $G(\omega)$ is the frequency-dependent GBK factor (below). The r^2 dependence is the direct quantum-mechanical trace of the dipole-coupling cross-section: a state with larger electronic extent couples more strongly to a passing electron.

Rotating to the SDT basis via the eigenvectors of $H_n(F, \mu)$ (Section 3.6) makes Φ_n approximately diagonal, since the dressed-state splitting $\omega_{kk'}$ is generally large compared to the collision rate. The diagonal element for upper dressed state $|k\rangle$ is:

$$\gamma_k^{(u)} = W_0 \langle k | r_u^2 | k \rangle [C_{n_u} + G(\Delta E_k)] \quad (51)$$

The resolvent $(\omega \hat{I} - L(F, \mu) - i\Phi(\omega))^{-1}$ then has eigenvalues $(\omega - \omega_k - i\gamma_k)^{-1}$ in the SDT basis, and Eq. (9) reduces directly to the Lorentzian sum of Eq. (68).

Shell-average approximation. Computing the per-SDT matrix element $\langle k | r_u^2 | k \rangle$ exactly requires rotating the r^2 operator to each dressed-state basis at every quadrature point (`electron_operator=True`). The default approximation replaces $\langle k | r_u^2 | k \rangle$ by the shell average $\langle r^2 \rangle_{n_u}$, giving a single homogeneous half-width:

$$\gamma_e(\Delta E) = W_0 \langle r^2 \rangle_{n_u} [C_{n_u} + G(\Delta E)] \quad (52)$$

The lower-manifold contribution $\gamma_k^{(l)}$ is retained in principle through Eq. (49); in practice, for Balmer lines ($n_l = 2$, $\langle r^2 \rangle_2 = 12 a_0^2$) it is a factor ~ 3 smaller than the $n_u = 3$ upper term and is dominated by the latter.

3.8.2 Density-Temperature Prefactor

$$W_0 = \frac{4\pi}{3} N_e \sqrt{\frac{2m_e}{\pi k_B T_e}} \left(\frac{\hbar}{m_e} \right)^2 \frac{\hbar}{e} \quad [\text{eV } a_0^{-2}] \quad (53)$$

The $\hbar/e$ factor converts rad s^{-1} to eV.

3.8.3 Shell-Averaged Mean-Square Radius

$$\langle r^2 \rangle_n = \frac{1}{n^2} \sum_{l=0}^{n-1} (2l+1) \frac{n^2}{2Z^2} [5n^2 + 1 - 3l(l+1)] a_0^2 \quad (54)$$

This expression scales as n^4/Z^2 and is *not* replaced by an approximate constant.

Caution: which $\langle r^2 \rangle$ PPPB intends is ambiguous, and unresolved as of this writing. Ferri et al. (2022) Eq. (19) write the broadening operator as $\vec{R} \cdot \vec{R}$ with " $\vec{R}$ the (emitter) electron position operator operating in the subspace of PQN n ." Projecting $\vec{R}$ onto the shell before squaring gives the *intra-shell* sum $\langle r_{\text{intra}}^2 \rangle_n$ (below) — the same operator ZEST uses, for the same $\Delta n = 0$ reasons given in the ZEST-justification paragraph below (Section 3.8.6) — rather than the full closure sum above. StarkZee keeps the full form as the `electron_model='pppb'` default only because the package's Ferri-figure comparisons were produced with it; the projected reading is available as `electron_model='pppb-intra'` (or `r2_form='intra'` passed directly to `electron_impact_width`). The two differ by $\times 2.44$ at $n = 2$ and $\times 1.89$ at $n = 3$. To be settled against digitized Ferri width data.

3.8.4 Strong-Collision Constant

Values from Ferri, Peyrusse & Calisti [1], Table 1:

$$C_n = \begin{cases} 1.50 & n \leq 2 \\ 1.00 & n = 3 \\ 0.75 & n = 4 \\ 0.50 & n = 5 \\ 0.40 & n \geq 6 \end{cases} \quad (55)$$

3.8.5 Dynamical GBK Factor

$$G(\Delta E) = \frac{1}{2} E_1(y(\Delta E)), \quad y(\Delta E) = \left(\frac{n^2}{2Z} \right)^2 \frac{\Delta E^2 + E_c^2}{\text{Ry}_\infty T_e} \quad (56)$$

where $\Delta E = E - E_{pk}$ [eV] is the detuning from the transition center, $E_c = \hbar\omega_c$ [eV] is the cutoff energy, $\text{Ry}_\infty \approx 13.606$ eV is the Rydberg energy ($e^2/2a_0$, the ionization energy of hydrogen), and $E_1(x)$ is the first-order exponential integral function defined for $x > 0$ by:

$$E_1(x) = \int_x^\infty \frac{e^{-t}}{t} dt \quad (57)$$

Cutoff frequency. The collision integral diverges logarithmically at large impact parameters unless truncated at a maximum ρ_{max} , or equivalently at a lower cutoff $\omega_c = v_{\text{th}}/\rho_{\text{max}}$ [8]. StarkZee follows the formulation of Ferri, Peyrusse & Calisti [1]:

$$\omega_c = \max(\omega_p, \omega_e, \omega_L, \omega_{\alpha\alpha'}) \quad (58)$$

- $\omega_p = \sqrt{N_e e^2 / (\epsilon_0 m_e)}$ — **electron plasma frequency**: Debye screening suppresses interactions at $\rho > \lambda_D = v_{\text{th}}/\omega_p$.

- $\omega_e = v_{\text{th}}/r_e$ — **configuration-change frequency**: the impact approximation requires each collision to be completed before the surrounding electron configuration changes appreciably. At high density or low T_e this can be shorter than the Debye timescale (see Section 3.8.6).
- $\omega_L = eB/m_e$ — **electron Larmor frequency**: in $\vec{B}$, electrons follow helical trajectories, reducing the effective interaction duration.
- $\omega_{\alpha\alpha'}$ — **emitter level-splitting frequency**: zero for hydrogen (degenerate l -subshells).

The largest frequency dominates, imposing the tightest bound on ρ_{max} .

Frequency-dependent vs. resonance-center width. When `frequency_dependent_width=True` (default), $\gamma_e(\Delta E)$ is evaluated **pointwise at the observation detuning from the gross-structure line center**, $\Delta E = E - E_0$, using the full $E_1(y(\Delta E))$ function — the PPPB convention $\Phi(\Delta\omega)$ of [1] Eq. (19), where $\Delta\omega$ is the detuning from line center. All Stark-dressed components share this single width function, so each component becomes non-Lorentzian: its far wings relax toward the strong-collision floor C_n as $G \rightarrow 0$, and a component sitting far from line center receives a reduced width at its own position. Setting this flag to `False` fixes the width at the line-center value $\gamma_e(0)$, which reduces computation time at the cost of accuracy in the far wings.

This single half-width $\gamma_e(\Delta E)$ is the only quantity the electron model hands to the profile builder; how it is applied to each Stark-dressed transition is described in Section 3.9.

3.8.6 ZEST electron broadening variant (`electron_model='zest'`)

Activating `electron_model='zest'` in `LineProfile.compute_profile` switches to the electron broadening model of the ZEST code [17]. The formula has the same structure as the default GBK expression but replaces the full shell-averaged $\langle r^2 \rangle_n$ with the *intra-shell* mean-square radius:

$$\gamma_e^{\text{ZEST}}(\Delta E) = W_{\text{pref}} \langle r_{\text{intra}}^2 \rangle_n [G_n + G_{\text{ZEST}}(\Delta E)] \quad (59)$$

Physical justification — $\Delta n = 0$ interaction channels. The ZEST paper (Section 2.1 of [17]) adopts two approximations that together restrict the electron broadening operator to within-shell matrix elements:

1. **No-quenching approximation**: perturbers may not induce transitions between the lower and upper manifolds participating in the radiator emission process.
2. **$\Delta n = 0$ interaction channels**: “only states belonging to the same Layzer complex [same principal quantum number n] may be mixed by Stark and/or Zeeman effects.”

A Layzer complex is the set of all n^2 states sharing principal quantum number n . These two restrictions reduce the formal sum over all intermediate states in the broadening operator to within-shell dipole matrix elements only, directly yielding $\langle r_{\text{intra}}^2 \rangle_n$ in place of the full $\langle r^2 \rangle_n$. The restriction is also automatic from the GBK dynamical factor: an inter-shell detuning (e.g. the $n = 3 \rightarrow n = 2$ gap of ≈ 1.89 eV at the conditions of Fig. 2) drives $G(\Delta\omega_{\text{inter}}) \rightarrow 0$ exponentially, so inter-shell contributions vanish regardless of the explicit approximation.

Intra-shell squared radius. Starting from the intra-shell radial element $\langle n, l | r | n, l \pm 1 \rangle = \frac{3n}{2Z} \sqrt{n^2 - (l \pm 1)^2}$ and angular weight factors $C(l, l+1) = (l+1)/(2l+1)$, $C(l, l-1) =$

$l/(2l+1)$, the per- l intra-shell sum is

$$r_{\text{intra},l}^2 = \frac{9n^2}{4Z^2} (n^2 - l(l+1) - 1) \quad [a_0^2] \quad (60)$$

This formula holds for all $l \in [0, n-1]$, including the boundary cases $l = 0$ (no $l-1$ neighbor) and $l = n-1$ (no $l+1$ neighbor). Averaging over the n^2 spatial states with $(2l+1)$ weight yields the exact closed form:

$$\langle r_{\text{intra}}^2 \rangle_n = \frac{9n^2(n^2-1)}{8Z^2} \quad [a_0^2] \quad (61)$$

Numerical values for H ($Z = 1$): $n = 2$: $13.5 a_0^2$; $n = 3$: $81 a_0^2$; $n = 4$: $270 a_0^2$. These are smaller than the full shell averages (33, 153, 468 a_0^2) by factors of 0.41, 0.53, and 0.58, respectively.

Maximum wave-number cutoff κ_m . The PPPB model fixes the minimum impact parameter as $\rho_{\min} = n^2 a_0 / (2Z)$ (Bohr orbit) and uses a combined cutoff $\omega_c = \max(\omega_p, \omega_e, \omega_L)$. The ZEST convention instead defines a temperature-dependent wave-number upper limit

$$\kappa_m = \min \left(\frac{Z}{n^2 a_0}, \frac{Z \sqrt{2m_e k_B T_e}}{\hbar n^2} \right) \quad (62)$$

The second branch is the inverse **Weisskopf radius** $\rho_W = n^2 \hbar / (Z m_e v_{\text{th}})$, $v_{\text{th}} = \sqrt{2k_B T_e / m_e}$ — the hydrogen strong-collision radius, *not* the thermal de Broglie length $\hbar / (m_e v)$. The geometric branch $Z / (n^2 a_0)$ dominates above $T_e \approx \text{Ry}_\infty \approx 13.6 \text{ eV}$; below that the Weisskopf branch takes over. The only cutoff frequency retained in all ZEST G-functions is the electron plasma frequency ω_p (no ω_e or ω_L terms), so B does not enter the ZEST broadening directly.

G-function choices (electron_model selector). Three alternatives are available for the dynamical factor $G(\Delta\omega)$ in Eq. (51), all sharing $\langle r_{\text{intra}}^2 \rangle_n$, the same $G_n = C_n$ strong-collision constants, and κ_m as defined above.

zest / zest-gbk (default ZEST). Closed-form exponential integral with a κ_m -derived argument:

$$G_{\text{GBK}}(\Delta\omega) = \frac{1}{2} E_1 \left(\frac{\Delta\omega^2 + \omega_p^2}{2 x^2 \omega_p^2} \right), \quad x = \kappa_m \lambda_D \quad (63)$$

This has the same E_1 structure as the PPPB formula but with $x = \kappa_m \lambda_D$ playing the role of $\rho_{\min} / \lambda_D$, and only ω_p in the additive term.

zest-lee. Analytical *min*-blend of the impact ($\Delta\omega \rightarrow 0$) and wing ($\Delta\omega \rightarrow \infty$) limits:

$$\begin{aligned} G_0 &= \frac{1}{2} \left[\ln(1+x^2) - \frac{x^2}{1+x^2} \right], \\ G_\infty(\Delta\omega) &= \frac{1}{2} E_1 \left(\frac{\Delta\omega^2}{2 x^2 \omega_p^2} \right), \\ G_{\text{Lee}}(\Delta\omega) &= \min(G_0, G_\infty(\Delta\omega)) \end{aligned} \quad (64)$$

G_0 (the line-center plateau) provides the plasma-cutoff floor; the E_1 wing term contains only $\Delta\omega^2$ (no $+\omega_p^2$) because the regularization is already captured by G_0 . At large detuning the two forms agree.

zest-dufty. Full random-phase-approximation (RPA) numerical integration accounting for Landau damping:

$$G_{\text{Dufty}}(\Delta\omega) = \int_{\kappa_{\min}}^{\kappa_m} \frac{e^{-x^2}}{\kappa |\varepsilon(\kappa, \Delta\omega)|^2} d\kappa, \quad x = \frac{|\Delta\omega|}{\kappa v_{\text{th}}}, \quad v_{\text{th}} = \sqrt{\frac{2k_B T_e}{m_e}} \quad (65)$$

where the longitudinal RPA dielectric function is

$$\varepsilon(\kappa, \Delta\omega) = 1 + \frac{1}{(\kappa \lambda_D)^2} \left[1 - 2x \mathcal{D}(x) + i\sqrt{\pi} x e^{-x^2} \right] \quad (66)$$

and $\mathcal{D}(x) = e^{-x^2} \int_0^x e^{t^2} dt$ is the Dawson function. The lower limit $\kappa_{\min} = |\Delta\omega|/(10 v_{\text{th}})$ removes the static ($\kappa \rightarrow 0$) divergence. This is the most accurate of the three models but evaluates one numerical quadrature per frequency point.

In practice the three G-functions give nearly identical line cores; differences appear in the far wings where Landau damping becomes significant.

3.9 Profile Accumulation

Each Stark-dressed transition ($p \rightarrow k, q$) is broadened by a Lorentzian of HWHM $\gamma_e(\Delta E)$:

$$L_{pk}(\Delta E) = \frac{1}{\pi} \frac{\gamma_e(\Delta E)}{\Delta E^2 + \gamma_e(\Delta E)^2} \quad (67)$$

The three polarization profiles are accumulated by vectorized summation over all microfield quadrature points and all SDTs:

$$P_q(E) = \sum_i W_{F,i} \sum_j W_{\mu,j} \sum_{p,k} S_q(p, k) L_{pk}(E - E_{pk}) \quad (68)$$

where $q=0 \rightarrow P_\pi$, $q=-1 \rightarrow P_{\sigma+}$ (blue), $q=+1 \rightarrow P_{\sigma-}$ (red). This direct summation realizes Eq. (10) in practice, with the frequency-dependent half-width $\gamma_e(\Delta E)$ of Section 3.8 evaluated at the detuning of each (E, E_{pk}) pair. The three accumulated channels are the polarization profiles $I_\pi, I_{\sigma+}, I_{\sigma-}$ entering the observed intensity of Eq. (11); for a line of sight at angle α to $\vec{B}$ they combine as $P_\pi \sin^2 \alpha + \frac{1}{2}(P_{\sigma+} + P_{\sigma-})(1 + \cos^2 \alpha)$.

3.10 Frequency Fluctuation Model (FFM)

The profile assembled in Section 3.9 treats the ionic microfield as frozen during emission. When the ions move appreciably on the emission timescale, that quasi-static average must be replaced by a dynamic one, and the FFM provides this without rebuilding the atomic calculation. Dynamic ion motion causes the static microfield to fluctuate over time; in the FFM [6] this is modeled as a stationary Markovian process that mixes the Stark-dressed transition components at rate ν_i .

The Stark-Zeeman Hamiltonian $H = H_A + V_E$ is still diagonalized at each field configuration (F, μ) , producing dressed-state frequencies ω_k and dipole weights $|d_k|^2$ — the Stark-Dressed Transitions (SDTs). This step is purely static: ω_k and $|d_k|^2$ depend only on the instantaneous ion field. Fast electron collisions add a homogeneous Lorentzian half-width γ_k to each SDT (the GBK width from Section 3.8), acting on a timescale short enough that the ion configuration does not change during a single collision. Ion dynamics

enter exclusively through v_i , estimated as the inverse time for an ion to cross the mean ion spacing at its thermal speed:

$$v_i = \frac{v_{th}}{r_i}, \quad v_{th} = \sqrt{\frac{2k_B T_i}{m_i}}, \quad r_i = \left(\frac{3}{4\pi N_i} \right)^{1/3} \quad (69)$$

The dynamic line profile is then given by the Sherman-Morrison form:

$$I(\omega) = \frac{r^2}{\pi} \text{Re} \frac{S(\omega)}{1 - v_i S(\omega)} \quad (70)$$

with the static propagator sum:

$$S(\omega) = \sum_k \frac{p_k}{v_i + \gamma_k + i(\omega - \omega_k)} \quad (71)$$

where $p_k = |d_k|^2 / r^2$ are the normalized SDT weights and $r^2 = \sum_k |d_k|^2$. The limit $v_i \rightarrow 0$ recovers the static profile; $v_i \rightarrow \infty$ gives a single Lorentzian (motional narrowing).

SDT binning (`sdt_bin_tol`). The microfield-and-angle quadrature loop typically produces thousands of SDTs per polarization channel, many at nearly the same frequency — the $S(\omega)$ sum above is $O(N)$ per frequency point, so this dominates the FFM runtime. Passing `sdt_bin_tol` (an energy tolerance in eV) merges every group of SDTs within one polarization channel whose frequencies fall in the same bin of that width *before* the Sherman-Morrison solve: intensities are summed and each merged frequency becomes the intensity-weighted mean of its group. Because the solver only ever sees the pair (p_k, ω_k) per SDT, this is exact in the limit `sdt_bin_tol` $\ll (v_i, \gamma_k)$ and reduces N by a factor of 10–100 in practice; `sdt_bin_tol=1e-5` (eV) is a safe default for typical Balmer-line conditions. Leaving it at its default of `None` performs the exact, unbinned calculation.

3.11 Thermal Doppler Broadening

Thermal motion of the radiating ions causes each photon frequency to be Doppler-shifted by $\delta\omega = \omega_0 v_z / c$. Averaging over a Maxwell–Boltzmann velocity distribution yields a Gaussian broadening with $1/e$ half-width

$$\Delta\omega_D = \omega_0 \sqrt{\frac{2k_B T_i}{m_{\text{ion}} c^2}} \quad \Longleftrightarrow \quad \Delta E_D = E_0 \sqrt{\frac{2T_i}{m_{\text{ion}} c^2 / e}} \quad (72)$$

where T_i is the ion temperature, m_{ion} the emitter mass, and E_0 the transition energy in eV. For H Balmer- α at $T_i = 5$ eV, this gives $\Delta E_D \approx 0.062$ meV — comparable to the Zeeman splitting at 1 T and negligible relative to the Stark width at $N_e \sim 10^{23} \text{ m}^{-3}$.

The two calculation paths apply Doppler broadening at different stages.

FFM path (`calculate_ffm_profile`). When `apply_doppler=True` (the default), Doppler broadening is applied *inside* `calculate_ffm_profile` on the energy grid, immediately after the Sherman-Morrison step. The convolution is a zero-padded FFT filter of length $2N$:

$$\tilde{P}_q(k) \longrightarrow \tilde{P}_q(k) \exp(-2\pi^2 \sigma_D^2 k^2), \quad \sigma_D = E_0 \sqrt{\frac{T_i}{m_{\text{ion}} c^2 / e}} = \frac{\Delta E_D}{\sqrt{2}} \quad (73)$$

where k is the discrete-frequency variable (cycles per eV) of the padded rfft and zero-padding to $2N$ avoids circular-wrap artifacts. Set `apply_doppler=False` to obtain the purely Stark-Zeeman FFM profile.

Static profile path (`calculate_static_profile`). `calculate_static_profile()` returns the purely Stark-Zeeman-broadened profile (quasi-static ion + electron impact only); Doppler is not included. It must be added afterward as a post-processing step via `convolutions.py`:

- `apply_doppler_broadening(wavelengths_nm, profile, Ti_ev, species='H')` — convolves with a Gaussian kernel of width $\Delta\lambda_D$.
- `apply_instrument_broadening(wavelengths_nm, profile, fwhm_nm)` — convolves with a Gaussian instrumental slit function.

Both operate on a *uniform wavelength grid* (convert energy grids to wavelength before calling) and use FFT convolution for efficiency. The Gaussian kernel is:

$$K_D(\lambda) = \exp\left(-\frac{(\lambda - \lambda_0)^2}{\Delta\lambda_D^2}\right) \quad (74)$$

The complete broadening pipeline for the static path is therefore:

$$I_{\text{obs}}(\lambda) = [I_{\text{SZ}}(\lambda) * G_D(\lambda)] * G_{\text{inst}}(\lambda) \quad (75)$$

where I_{SZ} is the Stark-Zeeman profile, G_D is the Doppler Gaussian, and G_{inst} the instrumental Gaussian.

4 Approximations and Numerical Choices

StarkZee is designed to be self-contained and fast enough for exploratory spectroscopy studies. The main approximations are explicit in the solver rather than hidden behind external atomic-structure packages:

1. **Hydrogenic basis and analytic radial functions.** Matrix elements are built from analytic hydrogenic wavefunctions [3],

$$R_{nl}(r) = \sqrt{\left(\frac{2Z}{n}\right)^3 \frac{(n-l-1)!}{2n(n+l)!}} e^{-Zr/n} \left(\frac{2Zr}{n}\right)^l L_{n-l-1}^{2l+1}\left(\frac{2Zr}{n}\right), \quad (76)$$

giving self-contained r and r^2 matrix elements without requiring Cowan, FAC, or another external structure code. Empirical NIST energies can still be injected for field-free level positions when accurate line centers matter.

2. **Within-shell Stark mixing.** The Stark operator is retained exactly inside each principal shell, but couplings to neighboring n manifolds are neglected. This is appropriate below the Inglis-Teller regime, where Stark shifts remain small compared with the shell spacing.
3. **Quasi-static ions with optional dynamics.** The baseline profile treats ionic microfields as static during the radiative event and averages over their distribution. When ion motion is important, the FFM layer modifies the static profile through a fluctuation rate rather than rebuilding the atomic calculation.
4. **Electron-impact broadening.** Fast electrons enter through a Lorentzian impact width. By default (`frequency_dependent_width=True`) the GBK width $\gamma_e(\Delta E)$ is evaluated at each component detuning. Setting `frequency_dependent_width=False` fixes the

width at the resonance-center value $\gamma_e(0)$, which is faster and usually adequate when far wings are not the observable of interest.

5. **Microfield orientation quadrature.** The angular integral is evaluated with an N_μ -point Gauss-Legendre rule over $\mu = \cos \theta \in [0, 1]$, using the symmetry of the Hamiltonian under $\mu \rightarrow -\mu$; see Section 3.7.

5 Built-in Reference Models

The `starkzee.models` package provides five independent lineshape models that share a common call signature and serve as benchmarks against StarkZee's fully coupled solver. They are grouped into tabulated models (reading precomputed databases) and analytical models.

5.1 Tabulated Models

5.1.1 Stehlé (MMM) — *stehle*

The Stehlé model reads precomputed Stark lineshapes from the Model Microfield Method (MMM) database [14] for hydrogen Balmer and Paschen transitions over a grid of N_e and T_e values. The tables contain the *unmagnetized* ($B = 0$) Stark + fine-structure profile as a function of a reduced detuning $\Delta\omega / F_0$ (Holtsmark normal field units). The pipeline is:

1. Bilinear interpolation over the (N_e, T_e) table grid (with a Fortran-style three-point hyperbolic interpolator for the detuning axis).
2. Thermal Doppler broadening via FFT convolution.
3. Normal Zeeman splitting applied *after* the table: the whole Stark+Doppler profile is rigidly copied to $\omega \pm \omega_L$ (Larmor frequency) with angle-dependent π/σ weights:

$$I(\theta) = \sin^2\theta I_\pi + \frac{1+\cos^2\theta}{2} (I_{\sigma+} + I_{\sigma-}) \quad (77)$$

The Stark and Zeeman effects are therefore treated as **separable**. This is valid only when the Zeeman splitting $\mu_B B$ is much smaller than the Stark width. Stehlé covers arbitrary (n_u, n_l) and species with no restriction on B -field value (it is simply ignored in the table lookup).

5.1.2 Rosato — *rosato*

The Rosato database [15] solves the Stark-Zeeman problem *jointly* and tabulates the result with B as an explicit table axis ($B \in \{0, 1, 2, 2.5, 3, 5\}$ T). The pipeline is:

1. Interpolation over the (N_e, T_e, B) grid. The B -interpolation is *scaled*: before blending two bracketing profiles at B_0 and B_1 , each profile's detuning axis is stretched by B_{node}/B . This exploits the fact that Zeeman splitting scales linearly with B , aligning the σ components before interpolating.
2. Angle dependence from *two real tables* (parallel and perpendicular to $\vec{B}$), blended as $I = I_\parallel \cos^2\theta + I_\perp \sin^2\theta$. Both tables carry the correct π/σ lineshape and width — the angle enters as a physical interpolation, not just an amplitude weight.
3. Thermal Doppler broadening via FFT convolution.

Rosato is restricted to deuterium Balmer lines and the table coverage $N_e \in [10^{13}, 10^{16}] \text{ cm}^{-3}$, $T_e \in [0.316, 31.6] \text{ eV}$, $B \leq 5 \text{ T}$.

5.2 Analytical Models

5.2.1 Lomanowski — *lomanowski*

Polynomial fits for the Stark FWHM from Lomanowski *et al.* [16]:

$$\Delta\lambda_S = c_{n_u n_l} N_e^{a_{n_u n_l}} T_e^{-b_{n_u n_l}} \quad [\text{nm}] \quad (78)$$

Coefficients (a, b, c) are tabulated for H/D Balmer and Paschen lines up to $n_u = 9$. A Voigt profile combines this Stark Lorentzian with a Doppler Gaussian. No magnetic field treatment.

5.2.2 Parameterized Stehlé — *stehle_param*

An analytical parametric fit to the Stehlé tables, providing a fast closed-form approximation to the field-free Stark profile without reading the NetCDF database.

5.2.3 Voigt — *voigt*

A pseudo-Voigt profile using the Griem α_{12} Stark half-width [7] combined with thermal Doppler broadening. Serves as a quick estimate when only order-of-magnitude Stark width accuracy is required.

5.3 Comparison with StarkZee

Feature	StarkZee	Stehlé	Rosato
Magnetic field	Full simultaneous diagonalization of $H = H_A + V_E$	$B = 0$ in tables; added as rigid triplet after	B is a table axis
Fine structure	Yes — spin-orbit, MV, Darwin	No (degenerate hydrogenic)	In tables
Electron broadening	GBK semi-classical, frequency-dependent	Unified theory (more rigorous far wings)	In tables
Ion dynamics	Quasi-static + optional FFM	Static + some ion-dynamics treatment	In tables
Microfield	Hooper screened	Holtmark/Hooper variant	Holtmark variant
B -treatment	Intrinsic to Hamiltonian	External convolution	B -scaled interpolation

The most important physical distinction at $B \neq 0$: StarkZee diagonalizes $H = H_A + V_E$ *simultaneously* for all field strengths. When $\mu_B B \sim 3n e a_0 F$ (Zeeman and Stark splitting comparable) the eigenstates are genuine Stark-Zeeman hybrids — neither pure Zeeman nor pure Stark states. No post-processing convolution can reproduce this mixing. The separable approximation (Stehlé) and scaled-interpolation approach (Rosato) both become inaccurate in this regime.

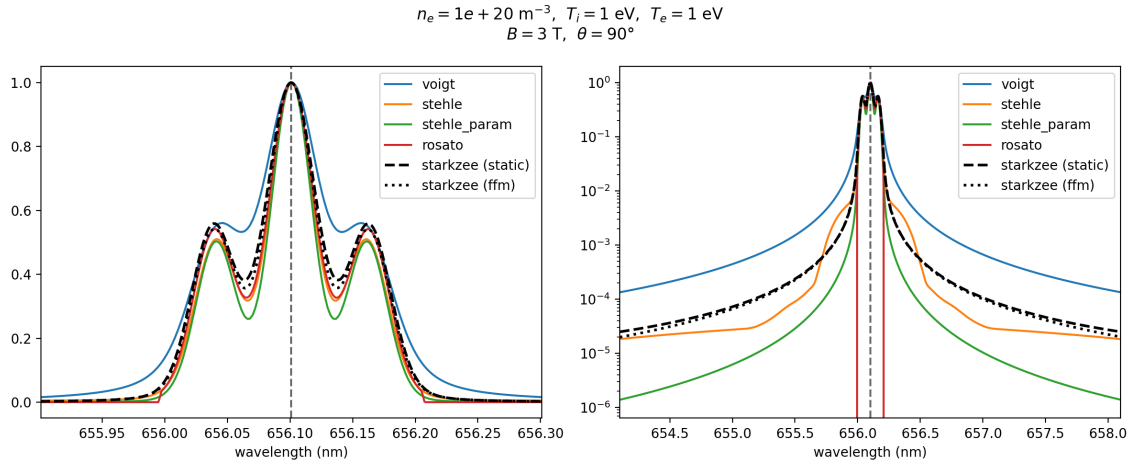


Figure 1: StarkZee (static and FFM) compared against four built-in reference models (Lomanowski has no magnetic-field treatment and is disabled in this comparison) for D_α ($n = 3 \rightarrow 2$) at $N_e = 10^{20} \text{ m}^{-3}$, $T_i = T_e = 1 \text{ eV}$, $B = 3 \text{ T}$, $\theta = 90^\circ$. Linear scale (left) and log scale (right). The field-free models (Voigt, Stehlé, Stehlé-parameterized) collapse the Zeeman triplet into a single unsplit line, while StarkZee and Rosato — which both treat B intrinsically — resolve the $\pi/\sigma^\pm$ splitting. Generated by `examples/model_comparison.py`.

A Physical Approximations and Domains of Validity

Approximation	Mathematical form	Domain of validity	Failure mode
Within-Shell Isolation ($\Delta n = 0$)	$V_E = \text{diag}_n(V_E)$, no $n \rightarrow n \pm 1$ coupling	Stark shift $\ll$ shell spacing $2Z^2\text{Ry}/n^3$	Quadratic Stark; Inglis-Teller merging at $N_e \gtrsim 10^{24} \text{ m}^{-3}$.
Quasi-Static Ion Microfields	$\vec{F}_{\text{ion}} = \text{const}$	Fluctuation rate $\nu_i \ll$ Stark width ΔE_S	Ion dynamics / motional narrowing; use FFM.
Semi-Classical Electron Impact	Lorentzian with GBK width γ_e	$\lambda_{dB} \ll$ impact parameter ρ	Quantum close-coupling needed for cold dense plasmas.
Dipole Approximation	$V_{\text{rad}} \propto \vec{r} \cdot \vec{E}$	$\lambda \gg \langle r \rangle_n \sim n^2 a_0 / Z$	Quadrupole/octupole for high- n Rydberg states.

Table 1: Key physical approximations, their domains of validity, and failure modes.

B Physics Explanation of Use Cases and Operational Limits

B.1 The High-Magnetic-Field Regime

In laboratory magnetic confinement fusion devices (tokamaks, stellarators), the magnetic field reaches $B \approx 1\text{--}10 \text{ T}$:

- The Zeeman splitting $\Delta E_Z \approx 0.1 \text{ meV}$ is comparable to the fine-structure splitting of lower shells, requiring the full coupled Hamiltonian diagonalization.
- Typical microfield strengths $F \sim 10^5\text{--}10^6 \text{ V/m}$ represent a weak-Stark regime where Stark broadening acts as a symmetric perturbation on the Zeeman triplet structure.

For astrophysical compact objects (magnetized white dwarfs with $B \sim 10^3\text{--}10^5 \text{ T}$, neutron stars with $B \sim 10^8 \text{ T}$):

- The quadratic Zeeman term $H_Z^{(2)} \propto B^2$ dominates, causing significant blue-shifting and highly asymmetric splitting. The `quadratic_zeeman=True` flag enables exact numerical treatment of the $\Delta l = \pm 2$ coupling (see Section 3.2).
- StarkZee's exact numerical computation of $\langle n, l_1 | r^2 | n, l_2 \rangle$ avoids the geometric-mean overestimation of up to 41% for $n = 5$.

B.2 Density Limits

- **Low density** ($N_e \lesssim 10^{18} \text{ m}^{-3}$): The profile is dominated by thermal Doppler broadening; microfield weights converge to a delta function at $F = 0$.
- **High density** ($N_e \gtrsim 10^{24} \text{ m}^{-3}$): The inter-particle spacing r_e approaches the atomic radius $\langle r \rangle_n$. The static and $\Delta n = 0$ approximations break down as the Stark shift exceeds the Rydberg shell spacing (Inglis-Teller limit).

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